Ahmedabad University
School of Engineering and Applied Science

Brain Tumor Segmentation and Localization
Using Traditional Image Processing and OpenCV

Weekly Status Report
Week 8 of 10

Course: ECE501 – Digital Image Processing — **Instructor:** Professor Mehul Raval — **Section:** 01
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1 Introduction

The main idea of this week's report is to explain the core idea behind the watershed segmentation which we are going to use and the basic differences between the previous method of thresholding.

2 Objective

- Compare the two methods concept wise to determine their similarities and differences.
- Document the OpenCV parameters of each method and how changing them affects the output.

3 Literature Review

Vincent and Soille formalised the watershed transform for digital images through an immersion (flooding) algorithm [8]. Otsu's method [9] remains the standard automatic global threshold selection method and underlies most threshold-based baselines. Gonzalez and Woods [10] describe both techniques as part of the segmentation module of the course syllabus.

For brain MRI, the works reviewed in Week 6 apply watershed in several forms: interactive multiscale watershed [1], marker-based watershed with contrast enhancement [2], gradient-based marker-controlled watershed using a fractional Sobel mask [3], morphological reconstruction with gradient modification to suppress spurious minima [4], morphological and marker-based watershed with thresholding [5], and marker-controlled watershed combined with fuzzy C-means [6].

4 Work Completed This Week

4.1 Methodology

Thresholding. Thresholding is a point operation: each pixel is classified using only its own intensity $f(x,y)$ and a threshold T,

$$g(x,y) = \begin{cases} 1, & f(x,y) > T \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

In global thresholding T is fixed. Otsu's method chooses T automatically by maximising the between-class variance of the histogram [9]. Adaptive thresholding computes $T(x,y)$ from the local neighbourhood of each pixel, which tolerates uneven illumination. In all cases the decision ignores spatial relationships between pixels, so the output can be noisy, fragmented, or contain several disconnected regions that morphology must later clean.

Watershed. The watershed transform treats a grayscale image as a topographic surface, where pixel intensity (or, more commonly, gradient magnitude) is the height. Flooding starts from the local minima, each minimum grows an individiul catchment basin, and where water from two different basins would merge, a border is built. These borders are the watershed lines and form the segmentation boundaries [8]. Using the gradient magnitude as the surface places the minima inside homogeneous regions and the ridges on object edges, so the dams settle on edges.

Marker-controlled watershed. Plain watershed starts a basin at every local minimum, so noise creates many tiny basins (over-segmentation). In the marker-controlled version, flooding is allowed to start only from user-defined or automatically generated markers: sure-foreground, sure-background, and an unknown band

between them that the algorithm resolves. This is why Week 7 uses thresholding and morphology first, to produce markers, and then watershed to place the boundary.

Core idea: similarities and differences.

Aspect	Thresholding	Watershed
Basic idea	Classify each pixel by intensity against T	Flood a surface from markers; boundaries form where basins meet
Uses spatial information	No, pixel-wise	Yes, through neighbourhood connectivity and gradient
Input landscape	Intensity histogram	Gradient magnitude or intensity surface
Output	Binary mask, boundaries may be open or fragmented	Labelled regions with closed boundaries
Needs markers	No	Yes (marker-controlled variant)
Typical failure	Similar intensity tissue is merged or missed	Over-segmentation, or leakage with poor markers
Main parameters	T, block size, constant C	Kernel size, iterations, distance-transform fraction, marker rule

Both methods partition the image into regions and both depend on a good intensity contrast between object and background. The difference is that thresholding answers "which side of T is this pixel?", while watershed answers "which marker's basin does this pixel belong to?". In the proposed pipeline thresholding is retained as a front end that supplies the markers.

4.2 Implementation Details

The snippets below are illustrative OpenCV implementations of the two methods (the watershed one follows the distance-transform workflow in the OpenCV documentation [7]) and show where the parameters enter. They are meant to explain the methods, not to report results.

Thresholding in OpenCV.

```
import cv2

gray = cv2.imread("slice.png", cv2.IMREAD_GRAYSCALE)
blur = cv2.GaussianBlur(gray, (5, 5), 0) # noise reduction

# 1) Global threshold: T is chosen manually
_, t_fixed = cv2.threshold(blur, 120, 255, cv2.THRESH_BINARY)

# 2) Otsu: T is chosen automatically from the histogram
T, t_otsu = cv2.threshold(blur, 0, 255,
                          cv2.THRESH_BINARY + cv2.THRESH_OTSU)

# 3) Adaptive: T(x,y) from a local Gaussian-weighted mean minus C
t_adapt = cv2.adaptiveThreshold(blur, 255,
                                cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
                                cv2.THRESH_BINARY,
                                31, # blockSize (odd)
                                -5) # C
```

Marker-controlled watershed in OpenCV.

```
import cv2
import numpy as np

gray = cv2.imread("slice.png", cv2.IMREAD_GRAYSCALE)
blur = cv2.GaussianBlur(gray, (5, 5), 0)

# Candidate region from thresholding
```

```
_, binary = cv2.threshold(blur, 0, 255,
                          cv2.THRESH_BINARY + cv2.THRESH_OTSU)

# Morphological clean-up
kernel = np.ones((3, 3), np.uint8)
opening = cv2.morphologyEx(binary, cv2.MORPH_OPEN, kernel,
                           iterations=2)

# Sure background, sure foreground, unknown band
sure_bg = cv2.dilate(opening, kernel, iterations=3)
dist = cv2.distanceTransform(opening, cv2.DIST_L2, 5)
_, sure_fg = cv2.threshold(dist, 0.5 * dist.max(), 255, 0)
sure_fg = np.uint8(sure_fg)
unknown = cv2.subtract(sure_bg, sure_fg)

# Marker image: 0 = unknown, 1 = background, 2.. = foreground labels
_, markers = cv2.connectedComponents(sure_fg)
markers = markers + 1
markers[unknown == 255] = 0

# Watershed needs a 3-channel image; boundary pixels become -1
color = cv2.cvtColor(gray, cv2.COLOR_GRAY2BGR)
markers = cv2.watershed(color, markers)
color[markers == -1] = (0, 0, 255)
```

Parameters and their expected effect.

Parameter	Method	Effect when increased
Threshold T (fixed)	Thresholding	Fewer pixels pass as foreground; small values include more tissue
`blockSize`, C	Adaptive	Larger block gives a smoother local estimate; C shifts the local threshold
Gaussian kernel size	Both	Stronger smoothing, but fine boundary detail is lost
Opening kernel, iterations	Watershed	Removes more small objects, may erode thin tumor parts
Dilation iterations	Watershed	Widens the sure-background boundary and shrinks the unknown band
Distance fraction (0.5 above)	Watershed	Smaller, more selective foreground markers; too high may miss part of the tumor

5 Results

The experiments will be done during the Diwali Holiday week and will be attached in the next report as this week's report was oriented towards not the results but implementation of the algorithm in the pipeline.

6 Challenges Faced

- **Over-segmentation:** noise and small intensity variations create spurious minima, which must be controlled through smoothing and markers.
- **Similar intensities:** The tumor and some healthy tissue overlap in intensity, which limits any intensity based marker generation.
- **Parameter dependence:** Thresholds, kernel sizes, and distance-transform fractions must be tuned across multiple slices, not a single example.
- **Baseline metric:** the threshold baseline is reported as accuracy, whereas overlap metrics such as Dice and IoU are more informative for small tumor regions, so both will be computed for a fair comparison.

7 Plan for Next Week

- Run the complete marker-controlled watershed pipeline on the selected BraTS 2020 slices.
- Inspect the sure-foreground, sure-background, and unknown markers visually.
- Test different threshold levels, kernel sizes, and distance-transform fractions.
- Compute Dice, IoU, precision, and recall against the ground-truth masks.
- Compare watershed against the existing threshold pipeline on the same slices.
- Analyse failure cases and identify refinements to reduce over-segmentation.

8 References

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